CALCULUS · UNIT 5: ANTIDERIVATIVES AND DEFINITE INTEGRALS

# Solar Energy Audit

*Teacher Guide & Answer Keys*

**Teacher**

2–3 class periods

## Project Overview

Students audit the school's solar panels. Power readings every two hours are summed with Riemann and trapezoidal sums to estimate energy, a power model is integrated with antiderivatives, and integral properties and u-substitution compare models.

Energy as accumulated power makes the area interpretation concrete.

## Standards Addressed

- AP LIM-5 Riemann sums and the definite integral
- AP FUN-6 Antiderivatives and evaluating integrals; u-substitution
- AP FUN-6.A Properties of definite integrals
- AP CHA-4 Accumulation in context

## Pacing (2–3 class periods)

| Day | Plan |
|---|---|
| Day 1 | Launch with a photo of solar panels and a meter. Part 1 sums. |
| Day 2 | Part 2 antiderivatives and integrals. |
| Day 3 | Part 3 models and the audit report. |

## Materials

- Student handout and Power Meter Readings
- Graphing calculators
- Graph paper
- Optional: real school energy data

## How the Three Versions Differ

| Level | What changes |
|---|---|
| CP2 | Left, right, and trapezoidal sums, basic antiderivatives, and one definite integral. |
| CP1 | Midpoint sums, over/underestimates, integral properties, and an upgrade inequality. |
| Honors | Error and concavity, limit of sums, sine model with substitution, and storage analysis. |

## Watch For (Common Misconceptions)

- Forgetting to multiply heights by Δt.
- Dropping + C on indefinite integrals.
- Reversing limits without changing the sign.

## Extension Ideas

- Compare with your home's electric bill.
- Model a cloudy afternoon with a piecewise function.
- Research how solar output depends on season.

# Answer Key – CP2

## Part 1: Estimate the Energy

| # | Task | Answer |
|---|---|---|
| 1 | Find the left Riemann sum for 6 a.m. to 6 p.m. | $2(0 + 24 + 58 + 72 + 60 + 30) = 488$ kWh |
| 2 | Find the right Riemann sum. | $2(24 + 58 + 72 + 60 + 30 + 4) = 496$ kWh |
| 3 | Find the trapezoidal sum (average of left and right). | 492 kWh |
| 4 | Is the school reaching its 600 kWh goal? By how much is it short? | No; about 108 kWh short |

## Part 2: Antiderivatives

| # | Task | Answer |
|---|---|---|
| 5 | $\int 6x^2\,dx$ | $2x^3 + C$ |
| 6 | $\int (4x - 3)\,dx$ | $2x^2 - 3x + C$ |
| 7 | $\int \cos x\,dx$ | $\sin x + C$ |
| 8 | Why do we add + C? | Every function that differs by a constant has the same derivative. |

## Part 3: Power Model

| # | Task | Answer |
|---|---|---|
| 9 | Find P(0), P(6), and P(12). | 0, 72, 0 |
| 10 | Evaluate $\int_0^{12} P(t)\,dt = \int_0^{12} (72 - 2(t - 6)^2)\,dt$. | $[72t - \frac{2}{3}(t - 6)^3]_0^{12} = 864 - 144 - (0 + 144) = 576$ kWh |
| 11 | Compare the model with your trapezoidal estimate and recommend whether to add panels. | 576 vs. 492; both are under 600, so add panels. |

## Reflection

| # | Prompt | Look for |
|---|---|---|
| R1 | Why does the area under a power graph give energy? | Energy = power × time; summing power × small time intervals gives the area. |
| R2 | Why are left and right sums different? | They use different heights for each interval. |

# Answer Key – CP1

## Part 1: Data Sums

| # | Task | Answer |
|---|---|---|
| 1 | Find the left, right, and trapezoidal sums. | Left 488, right 496, trapezoid 492 kWh |
| 2 | Use a midpoint sum with 3 intervals of 4 hours (midpoints 8 a.m., 12 p.m., 4 p.m.). | $4(24 + 72 + 30) = 504$ kWh |
| 3 | Power increases from 6 a.m. to noon. On that interval, is the left sum an over- or under-estimate? Why? | Underestimate; left endpoints are the lowest heights on increasing intervals. |
| 4 | Write the energy from 6 a.m. to 6 p.m. as a definite integral using P(t). | $\int_0^{12} P(t)\,dt$ |

## Part 2: Antiderivatives & Integrals

| # | Task | Answer |
|---|---|---|
| 5 | Expand P(t) and find an antiderivative. | $P = -2t^2 + 24t$; $F(t) = -\frac{2}{3}t^3 + 12t^2$ |
| 6 | Find the energy from 6 a.m. to 12 p.m. and from 12 p.m. to 6 p.m. | Each 288 kWh (by symmetry) |
| 7 | Use properties to find $\int_0^{12} P(t)\,dt$ from the two halves. | 576 kWh |
| 8 | Find $\int_{12}^0 P(t)\,dt$ and explain the sign. | −576; reversing the limits changes the sign |

## Part 3: Upgrade Plan

| # | Task | Answer |
|---|---|---|
| 9 | If each new panel adds 4% to P(t), how many panels are needed so $\int_0^{12} P\,dt \geq 600$? | $576(1 + 0.04n) \geq 600 \rightarrow n \geq 1.04 \rightarrow 2$ panels |
| 10 | A cloudy day has power 0.6P(t). Find the energy and the shortfall. | 345.6 kWh; 254.4 kWh short |
| 11 | Write the audit report with a graph, sums, and integrals. | Answers vary. |

## Reflection

| # | Prompt | Look for |
|---|---|---|
| R1 | How do Riemann sums connect to definite integrals? | The definite integral is the limit of Riemann sums as the intervals shrink. |
| R2 | Why is the midpoint sum often more accurate? | Midpoint heights balance overestimates and underestimates within each interval. |

# Answer Key – Honors

## Part 1: Error Analysis

| # | Task | Answer |
|---|---|---|
| 1 | Compute the trapezoidal sum with the data and with the model values at t = 0, 2, …, 12. | Data 492; model values 0, 40, 64, 72, 64, 40, 0 → 2(40 + 64 + 72 + 64 + 40) = 560 |
| 2 | Find the exact integral of the model and the error of the model's trapezoidal sum. Why is it an underestimate? | 576; error 16; P is concave down so trapezoids lie below the curve. |
| 3 | Write the limit of a right Riemann sum that equals $\int_0^{12} (-2t^2 + 24t)\,dt$. | $\lim_{n\to\infty} \sum_{k=1}^{n} \left(-2\left(\frac{12k}{n}\right)^2 + 24\left(\frac{12k}{n}\right)\right) \cdot \frac{12}{n}$ |
| 4 | Which data reading disagrees most with the model? What might explain it? | At 8 a.m. (24 vs. 40); morning shade or haze |

## Part 2: Better Models

| # | Task | Answer |
|---|---|---|
| 5 | Find $\int_0^{12} 72 \sin\left(\frac{\pi t}{12}\right) dt$ using substitution. | $u = \frac{\pi t}{12}$: $\frac{864}{\pi}[-\cos u]_0^{\pi} = \frac{1{,}728}{\pi} \approx 550$ kWh |
| 6 | Compare Q with the data at 8 a.m. and 4 p.m. Which model fits better? | Q(2) = 36, Q(10) = 36 vs. data 24 and 30; Q fits the data better than P. |
| 7 | Find $\int t\, e^{(-t^2)}\,dt$ using u-substitution and explain each step. | $u = -t^2$, $du = -2t\,dt \rightarrow -\frac{1}{2} e^{(-t^2)} + C$ |
| 8 | Find the average power over the day using Q. | $\frac{550}{12} \approx 45.8$ kW |

## Part 3: Upgrade Analysis

| # | Task | Answer |
|---|---|---|
| 9 | How many 4% panels are needed so the energy reaches 600 kWh? | 550(1 + 0.04n) ≥ 600 → n ≥ 2.27 → 3 panels |
| 10 | Battery storage starts charging when power exceeds 40 kW. Find when Q(t) > 40 and the energy above 40 kW. | $\sin\left(\frac{\pi t}{12}\right) > \frac{5}{9}$ → 2.25 < t < 9.75 (about 8:15 a.m. to 3:45 p.m.); ∫ from 2.25 to 9.75 of (Q(t) − 40) dt ≈ 157 kWh (technology or antiderivative) |
| 11 | Write the audit report comparing the models, with error analysis and your recommendation. | Answers vary. |

## Reflection

| # | Prompt | Look for |
|---|---|---|
| R1 | How does concavity tell you whether a trapezoidal sum over- or underestimates? | Concave down: trapezoids are below the curve (under). Concave up: above (over). |
| R2 | Why choose one model over another when both are reasonable? | Compare how well each fits the data and whether its shape matches the physical situation. |